
HEPHAËSTUS
OR
The Soul of the Machine

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E. E. FOURNIER D'ALBE

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E. E. FOURNIER D'ALBE

HEPHÆSTUS

HEPHAESTUS
OR
The Soul of the Machine

BY
E. E. FOURNIER D'ALBE
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PREFACE

The purpose of this little book is to show that one, at least, of the gods of Hellas has survived the flood which swept away the most entertaining company of gods and goddesses ever created by man's imagination. As I propose to set him in the august place vacated by the death of Zeus, a few biographical details may not be out of place.

Hephæstus was the son of Hera (Juno), but not of Zeus (Jupiter). How his mother put him into the world is not precisely known. Neither Zeus nor any other male god had anything to do with it. Yet it would be inappropriate in the case of such a confirmed matron as Hera to speak of parthenogenesis. Some extraordinary event had to take place before the great home-goddess could be driven to spite her lord and master by producing a son without his co-operation. And such an event had indeed occurred, for Zeus had suddenly reverted to one of the oldest forms of propagation known to biology, viz., propagation by budding. A fully-armed young goddess, severe of countenance and lithe of limb, had sprung forth from his head, thenceforth and for ever to lead and dominate the world of thought. It was up to Hera to match Pallas Athene by some equally important contribution to the evolution of gods, and so by some mysterious process, into which Greek historians did not care to pry, she produced Hephæstus, whom the Romans called Vulcan, now the only surviving representative of that lively and enterprising clan which once ruled the world from the summit of Mount Olympus.

Like many another product of inspiration, Hephæstus was at first regarded as a failure. He was undersized and weak-chested, and Hera had to suffer much from the gibes of her peers and peeresses. So one day she dropped him down the slope of the mountain and he fell into the sea. He was picked up by two of those charming and motherly sea-goddesses which at that time abounded in sea-water, and was brought up in a grotto under the ocean. In return for their kindnesses he made them pretty ornaments of coral and mother-of-pearl.

At the age of ten, or thereabouts, he set out to find his mother. It was some time before Hera recognized in the lame boy, with the spinal curvature and the swarthy but pleasant face, the child she had so mercilessly cast off. But a spark of mother-instinct revived under the flame of the child's filial devotion, and they soon became lasting friends and allies.

Hephæstus, naturally, owed no allegiance to Zeus, and in the frequent marital disputes between him and Hera he invariably took his mother's part, and so successfully that the redoubtable Father of the Gods took him by his lame leg and flung him into space. He fell for a whole day^[1] and eventually alighted, like a meteorite, on the island of Lemnos, where he was worshipped as many a meteorite has been worshipped before and since. He put up with this for a while, but the blood of the Olympians asserted itself, and he painfully climbed home once more. This time he succeeded in planting his unequal feet firmly on his native rock, and he soon became a favourite among his divine relations. He undertook the reconstruction of the Olympian dwellings, and for this purpose established a wonderful workshop with a huge anvil and twenty bellows, all of which would work at his mere behest. The workshop was made of fire-proof materials, and shone "like a star in the night."

^[1] This must be a mistake, as it would put the summit of Olympus somewhere beyond the orbit of the moon.

He solved the housing question by building a separate palace for each of the gods and goddesses, replete with all the comforts and refinements of civilization. Nor did his amiability end there. Sometimes, at their daily assemblies, he would give Ganymede a day off and would himself hand round the nectar, producing roars of laughter by the contrast between his hobbling gait and the deportment of the graceful young cup-bearer.

He became so popular that at last he was able to marry Aphrodite herself, the goddess of Love and Beauty, whom the Romans called Venus, and who was born of the foam of the sea somewhere off Cyprus. But this union of Beauty and the Beast was far from being a happy one. Venus was not satisfied with a "lame mechanic" as she called him, and hankered after the dashing Ares, alias Mars, the god of war. Having been warned by Helios, the sun-god, of the progress of the intrigue—after it had already reached its climax—Hephæstus lay in wait for the guilty pair with an invisible net, and having caught them, dragged them before the other gods amid Homeric

laughter produced by their struggles in the invisible meshes. And when a daughter was born, whom they miscalled Harmonia, Hephæstus had his revenge by presenting her with a necklace which brought disaster to her and all its later possessors until it was finally laid to rest in the temple of Athene.

Meanwhile, the fame of his works spread far and wide. King Aetes of Colchis ordered the construction of two bronze bulls capable of breathing fire through their nostrils, and when Achilles, the Greek champion before Troy, decided to kill Hector and avenge his friend Patroclus, he went to Hephæstus with an introduction from Thetis, the sea-goddess, and prevailed upon him to make him a marvellous shield on which all heaven and earth were figured in bronze, and which was quite impervious to mortal spear and battle-axe.

And so he dwelt and worked on Olympus for three thousand years or so, establishing branch works on Lipari and in Sicily which kept working at full blast until Paul of Tarsus came with his claim to have found the Unknown God who was to establish a new Roman dominion to take the place of the mighty Empire of the Cæsars, and was incidentally to sweep away the gods of Rome and Greece alike and establish the worship of a tripartite God who never smiled.

So the light-hearted company of Mount Olympus died, all but Hephæstus, who hobbled through many lands seeking a place where he might work and set his bellows blowing once more. For centuries he wandered, despised and jeered at by monks, hermits, and anchorites. But his courage never failed him. It survived the desolation of the Olympian palaces as it had survived his two falls from the summit.

And now he has come back into his own, and his power is spreading like a conflagration. The God of Fire is the supreme master of the earth. His furnaces are roaring. He has dispelled the clouds of Asiatic mysticism which obscured his native mountain. He has girdled the world with hoops of steel. He has found strength

“To break this sorry scheme of things entire,
And mould it nearer to the heart’s desire.”

And his victorious progress is only beginning to take effect. How it has come about, and what will be the eventual result of his domination of our planet, it is the purpose of what follows to elucidate.

HEPHÆSTUS

I

Ethnologists believe that the metamorphosis from beast-like savage to cultured civilian may be proximately explained as the result of accumulated changes that found their initial impulses in a half-a-dozen or so of practical inventions.

An Irish poet, disconsolately walking along Fleet Street one day, bethought himself of a small island in an Irish lake, where he could escape from the noise and bustle of the London streets. He wrote a lovely poem embodying that thought, a poem expressing the longing of every sensitive soul to retire within itself for a time and reconstruct its world from within instead of having it impressed from without.

Needless to say, he did not content himself with planting nine beans in a row, as he had longed to do in his fit of depression. On the contrary. He became an Irish statesman and Senator. He dwelt in surroundings far removed from the Stone Age. The very chair on which he sat down to write owed its existence to many factories and mechanical processes. Its wood was derived from planks cut in the saw mills. Its French polish, although put on by hand, was made of constituents—shellac, methylated spirit, whitening and what not—drawn from many sources, each involving a number of ingenious inventions. The coal which made his fire had been brought across the sea by a steamer—a miracle of complicated mechanism—and the shovel with which he put it on had been mined in Belgium, smelted in Germany, rolled in Sheffield, and shaped and finished in Birmingham.

Not that “Inishfree” was but an idle dream. There are many such islands in Ireland, peopled by men and women and children, living in one-roomed mud hovels on potatoes and stewed tea, infested with vermin and ravaged by tuberculosis and other infectious diseases, people who, if they cannot

emigrate, suffer their fate with a stolid and pathetic resignation relieved by the hope of heavenly compensation hereafter.

Such is Inishfree, stripped of its glamour and shown in its naked reality.

Savagery does not mean simplicity but complexity, not peace but constant dread, not health but hopeless disease and premature death. The South American Indian suffers acutely from constipation, which is about the last disease we should expect him to suffer from. The negro in his native Africa is often horribly deformed through neglect at birth, neglected injuries, or the ravages of insect pests.

It has taken mankind ten thousand years to emerge from savagery into barbarism, and from barbarism into civilization. Nor is the process yet completed. The mass of mankind is still grossly ignorant. But the leaven of knowledge is steadily working and pervading the mass, and the rate of progress is constantly rising.

Let us trace this wonderful process of civilization to its origins and probe its essential nature.

Starting somewhere in the tropics as a diminutive “sport” allied to the arboreal ape, man learned to use tools and weapons. This was the first step taken towards the conquest of the animal world, his natural enemies.

We do not know how many ages elapsed before the rough stone hammer evolved into the axe of polished flint. In any case, the rough stone, even if only used for throwing at his enemy or killing his prey, was an essentially new departure in evolution, and sounded the keynote to all that followed down to our own times.

Tools and weapons had been evolved by other animals, but they were always organically connected with their bodies. The tiger’s claws and teeth, the tusks of the elephant and the wild boar, the beak of the eagle were all formidable weapons and useful tools, but they could neither be detached, nor replaced, nor exchanged for other tools more suitable to the occasion. Those “natural” tools and weapons had all been produced by “evolution,” in other words, by a mysterious agency which some call God, some Adaptation, and some the Urge of Life.

Who was that audacious man who first took upon himself the divine privilege of making tools and weapons for himself, instead of waiting for

“nature” to provide them for him?

Perhaps he was a puny boy, lame from birth and unable to escape from the boar who was pursuing him. In the extremity of his terror he took up a heavy stone from the ground and flung it at the boar with all his might. The boar, we may well imagine, was dazed and probably terrified by this unusual method of defence, and slunk away from his feeble but resourceful antagonist. The story of that deliverance may, for all we know, have been sung for many generations in the tribe, and who knows but that the boy grew up into a great red-haired man, as great an adept at stone-throwing as the street-urchins in Belfast, and formed the prototype of the God Thor, whose hammer, when thrown, returned to his hand of its own accord.

But the essential step was taken. The lad had no teeth or claws which could match the solid tusks wielded by his mighty antagonist. So he “invented” a new weapon. He did not grow it in his own organism, as the squid grows its ink, to be used on one occasion and then gradually renewed. He took a piece of the outside world and made it temporarily a part of his person, a part which could be detached and resumed at a moment’s notice, a temporary attachment or extension of his body which required no blood-vessels to keep it in repair and which, if broken or injured, would inflict no pain upon himself.

This great innovation may have been aided by some analogies in the animal world. A bird had to collect twigs and leaves in order to build a nest, and had thus to put a portion of the “outside world” to its own uses. Besides, the very process of eating involved the apprehension of outside objects, an apprehension which, in the case of the lowest form of animal life, is to this day accomplished by pouring its jelly-like body round the object to be consumed.

But the next invention, the discovery of the use of fire, was a departure without a parallel to anything in the kingdoms of Nature. It placed mankind by a single act in a position of god-like authority over the living world. It is not surprising that the discovery of fire is surrounded by countless legends. In Greek mythology the first use of fire is attributed to Prometheus, the Fore-Thinker, the man who thinks ahead. He stole it from the heavens, and was punished by Zeus with terrible torment for having dared to endow mankind with a divine privilege.

How great the privilege was we cannot even yet realize. For the uses of fire are by no means exhausted, and are multiplying from day to day. But the essential element of the change was that something came into the hands of man which did not exist in "Nature" at all, not, at least, in a manner ordinarily accessible to organic being. It was quite "unnatural" to use fire. Fire represented a state in which no organism could survive, in which all its functions were ruthlessly stopped, and its living tissues destroyed. No animal except man has ever attained to the use of fire. Its use represents the transcendence of man from the ordinary scheme of Nature and his ascent into a sort of supernatural, or at least super-organic realm.

From this step there has been no retreat, and there can be no retreat until the sun itself grows cold. Whatever our "back-to-Nature" cranks may say, mankind cannot repudiate and renounce its most precious acquisition and all that it involves in the present and the future. We may dislike the smoke of blast-furnaces, but the remedy is not to do away with them, but to stop their smoking. The bellows of Hephæstus are blowing and his fires are burning. The age of machinery, begun in far-off Palæolithic days, but only established within the last hundred years or so, has now gripped us in a scheme from which none may escape.

Taken in its narrower sense, the word "machine" means a contrivance for increasing the force we can bring to bear upon objects until it exceeds the limits imposed upon the tension of our muscles. As most of the 400 voluntary muscles of the human body are attached to the bones in a manner which diminishes the force exerted below the actual tension of the muscles, the adoption of machinery constitutes a reversal of the "natural" use of the system of levers which we call our skeleton. The Lever, the Inclined Plane, the Wedge, the Pulley, the Wheel and Axle, and the Screw, are all contrivances for slowing down the rate of work until the force required to perform it comes within the compass of our muscles. Our bones, on the other hand, are mostly levers "of the third order," in which the force is applied between the fulcrum and the object to be moved. Thus it happens that in order to lift a weight of one pound, our muscles are strained by a force which may amount to as much as six pounds.

We are hardly justified in ascribing this uneconomic arrangement to the "stupidity" of Nature. Any other arrangement, such as the use of levers of the "second order," would have involved a much bulkier and clumsier build

of the muscular system. Nor can we conceive of the adoption by man of the Six Machines as a deliberate imitation of Nature, or an improvement upon observed natural processes.

The Six Machines are obviously the crude results of long-continued experience, and it was not until the middle of the nineteenth century that their real significance in the system of mechanics was recognized.

The real importance of these contrivances lies in the fact that they increase the power of the human body without any further evolution of the human organism. The adoption of mechanism was a turning point in evolution, and an event the significance of which can only be classed with such far-reaching innovations as the birth of life itself or that adoption of locomotion as an aid to nutrition which gave rise to the animal kingdom.

The most widely useful of the Six Machines was undoubtedly the Wedge. Its use was usually accompanied by that of the projected stone. The latter, after being swung by muscular force, was a reservoir of energy, which is the power of performing work. If brought to rest within a shorter distance than the curve of the swing, it required a correspondingly greater force to stop it. And that force could be still further increased by applying it to the end of a wedge whose sides bore upon the object to be split. The combination of mass and wedge is represented by the spear, the arrow, and the sword, while the hammer and the club acted as crushers rather than cutters.

Observe, then, this puny but formidable creature emerging from his tropical forest. In his hand he carried the means of annihilating that continuity of the organism which is essential to its existence. He was able to drive a powerful and irresistible wedge into the body of his antagonist and thus end the co-ordination of its natural functions. It was thus that, later in history, the Macedonian phalanx split up the armies of the Persian kings. By a different but essentially similar process the 15-inch naval gun drives a wedge clean through the armour of its opponent, in the shape of a shell which does not explode until it is right inside the enemy ship.

Man's war of conquest against the animal world had begun. He brought to them a death more sudden than had existed up to his time. For he had the power of driving his wedge into the most vulnerable and essential part of his enemy's body. A clean cut inflicted at the appropriate point would mean his final end. Man was not long in discovering his enemy's weaknesses.

And when his enemy was a human one, he made assurance complete by bringing home his head.^[2]

[2] It is curious in this connection that among the legends of the Saints there is no record of a decapitated person being brought back to life. The angels or departed saints who came to heal the torn breasts of martyred Christian virgins were unable to put heads back on bleeding trunks, even though the tongues in some of those heads were empowered to testify for some time after they would, in the ordinary course of things, have become silent for ever.

If we examine the connection between man and his weapon we find that the latter differs from an eagle's beak or a tiger's claw in but one essential point: it is no longer an integral part of the organism. It can be detached at will, and replaced by another weapon. Its separation or destruction does not imply an injury to the organism. Man is not put out of action by losing his weapon. He is only reduced to his original position with respect to his antagonist. And even that is no irreparable loss so long as he has other weapons at his command. Thus the same step which vastly increased his offensive power also made him comparatively immune to attack.

Now, a weapon in a man's hand, so long as it is in active use as a weapon, is a part of the man himself. It is true that he can lose it without perishing himself, but he can also lose an arm or a leg and still survive. The mere fact that the man's blood circulates in his natural leg and not in a wooden leg he may substitute for it makes no essential difference. He may kick with either. And we know that a man's leg, like all the cells of his body, is largely compound of inert matter such as food products and waste products, besides being nine-tenths water—an inorganic substance. A wooden leg, or any weapon which a man may use, may therefore be regarded as a limb of the man's body, so long, that is, as it is in active use. And if a "soul" animates that man's body and drives it to perform deeds of valour, the same soul will animate his weapon. The soul of the weapon is the soul of the man who uses it.

There is an increasing tendency in modern thought to abolish the distinction between soul and body and to regard them as one and indivisible. Adopting that view, we may assert that the use of a weapon means the enlargement of a man's body and the simultaneous expansion of his soul. Every weapon, every tool, every machine is the embodiment of a human thought and purpose. The user adopts that thought and purpose, and behold—the machine has found its soul!

II

Man is an animal who laughs and cooks.

When Prometheus had caught some sparks from the chariot of the sun and brought them safely to earth, hidden in a tube, there was consternation on Mount Olympus. The Conservative Government of that privileged stronghold trembled for its celestial prerogatives. It was all very well, they said, for Hephæstus to blow his bellows in his workshop and produce beautiful things for Olympians, so long as he did it under proper supervision. But fire, once brought to earth, would set the whole world ablaze and consume them in their palaces.

Hephæstus himself was not perturbed. Remembering the kindness he met with on Lemnos after his brutal expulsion from the company of the gods, he secretly sympathized with Prometheus and his race, and planned a closer co-operation with mankind. His Sicilian workshops saw him oftener than before, hobbling about among his furnaces and experimenting with every kind of ore and ingot. Wherever he went, whether among gods or men, his kindness and his merry humour made him a general favourite.

Although he often made weapons, he much preferred to make tools and ornaments. Among the Olympians, his strangely begotten “sister” Athene was his closest ally. Like him, she favoured the arts of peace rather than those of war, and despised the noisy and swashbuckling Ares. When Athene went to take up her abode in her Parthenon at Athens, he forsook Olympus and removed his furnaces to Lipari, off the Sicilian coast. And it was in Sicily, some centuries later, that his great disciple Archimedes was born, the man who was fitted and destined to establish the reign of Hephæstus in spite of the strenuous opposition of the Roman republic and its subsequent imperial and ecclesiastical successors.

When man was endowed with fire he received the gift without knowing whence it came or what it meant. Greek mythology, with its deep insight into unseen things, presents us Helios, the sun-god, as a friend of Hephæstus, the god of fire. We moderns express the same truth differently.

We say that all fire on earth is ultimately derived from the sun. Archimedes kindled fires—preferably in Roman ships—by means of concave mirrors which condensed the heat of the sun to a focus. The fuel for the fires was also provided by the heat of the sun, absorbed by the chlorophyll of the leaves in the forest and used to build up combustible wood and other products. The firing of the wood reversed the process of accumulation of solar energy and provided light and heat by day and by night, independently of the sun, the original giver. For a hundred million years already had the sun shone on a habitable earth, pouring out its light and heat and nourishing the luxuriant vegetation which covered most of the globe. For untold ages had the plants stored up the sunlight for some unknown end. That end became manifest when Prometheus kindled the first fire and made a new realm accessible to man. Till then, mankind had undergone the annual and daily vicissitudes of heat and cold due to the days and the seasons, but he had not controlled them. When he acquired the mastery of fire, he was enabled to wander north and south and take with him in his brazier the accustomed heat of the tropics.

But there was more to come. The invention of pottery—another great epoch—enabled him to create a solar furnace on a small scale. It enabled him to create a miniature world where the heat was much greater than in his own world. He soon found that plants and animals, passed through this new world, became richer and more palatable, and in many ways better adapted to his digestive system. It must have seemed highly “unnatural” at first to accelerate the ripening of fruits and predigest animal food by plunging them into water in which no living thing could survive, but the departure from the habits of the animal world, once begun, was never arrested, and it is still proceeding in our own day.

Our remote progenitors must have looked upon the guardians of their fires with much the same awe as that which mediæval people felt for sorcerers and alchemists, and with rather more justification. For man had everything to learn about the new power placed in his hands. He was beginning to find his way about a new world in which the ordinary laws of Nature were suspended. Water was no longer cold and wet in that world. It was hot as the sun and thin as smoke. It no longer flowed but rose in the air. Salt rapidly became invisible in it, but could as rapidly be recovered by

adding more fire. There was as much to learn about fire as there was about X-rays in 1895 and about “atmospherics” to-day.

The possession of fire turned winter into summer and night into day. It lengthened the life-time of its fortunate possessors. But it did much more than that. It endowed mankind with a number of gifts which must, in his primitive condition, have appeared to him as supernatural. It gave him substances which combined the hardness of stone with the toughness of wood. It enabled him to mould these substances into any desired shape by the softening action of excessive heat. Hephæstus himself cast his products in bronze, having cunningly mixed copper with tin in order to produce an alloy combining hardness with ductility. But he and his later disciples found the perfect substance in iron, which, though requiring an extreme degree of heat, yielded weapons and tools of unsurpassed power and strength.

And thus it was that Hephæstus and his human allies prepared that career of conquest which eventually swept over the earth and made all things new. The unprogressive Zeus, content with having learnt how to shake Olympus with a nod of his head, perished in the flood of new ideas from the East. Hephæstus, alone of all that shining company, had established a firm footing in the world. For the work of his hands had trained his brain and enabled it to build up a scheme of the cosmos in which every detail corresponded to some ascertained reality. And so, while Egyptian priests were speculating about the weight of a soul, while Hindu sects were vying with each other in producing the most hideous idols, and while Christian theologians were endeavouring to prove that three ones make one, Hephæstus and his followers were engaged in founding that superb edifice of knowledge which was destined to outlast the fall of empires and the dark ages which followed the eclipse of classical learning.

But even as late as 1800 A.D., in spite of the mariner's compass, the printing press, and gunpowder, the ground won by Hephæstus was very small. The postal services in Europe were inferior to that of the Cæsars, there was no telegraphic system equal to that of the ancient Persians, and the most stately sailing ship barely excelled the classical trireme. The fruits of the work of Newcomen and Watt and Papin had not yet blossomed. When they did come forth, things happened swiftly. Fire was made to generate steam, and steam, in the course of a century, transformed the world. The motive-power being there, it was put to a million uses. The

Wheel became the most important of mechanical contrivances. Now a wheel is a thing which, in the organic world, is simply impossible. In the days when microscopes were of feeble power, it was thought that the “wheel animalcule,” called *Vorticella*, had small wheels revolving on the rim of its opening to gather in the floating food particles. But higher magnifying power showed that the appearance was an illusion created by the oscillation of the fine hairs fixed round the aperture. A wheel revolving on a shaft implies a separation which precludes all organic connection. Had it been otherwise, an animal running on its own wheels would have made its appearance on earth long ago. The shaft of the steam engine could be provided with numerous pulleys, each capable of driving a machine connected with it by a belt. Then arose that marvellous crop of mechanical combinations which brought about the age of mechanism, combinations which, in the last resort, can be reduced to sliding couples and turning couples, and can be catalogued by means of formulæ, like words in a dictionary. So numerous were these inventions, so widespread was their use, and so great their effect on the minds of those generations, that a great illusion arose which coloured the philosophies of a whole century and has left traces even in modern times. People became so accustomed to the perfect functioning of a machine or a clock, so impressed with its regularity and intelligible complexity, that they began to look upon mechanism as a primary thing capable of “explaining” many non-mechanical things, not excluding the phenomena of life. Why should not man himself, the inventor of machinery, be a machine? If any functions of his body seemed beyond explanation on mechanical lines, might this not be solely due to their great complexity?

In a recent article in *Nature*, Professor Fraser Harris puts the matter tersely as follows:

“Because the stomach ‘works’ rhythmically and predictably we may call it a machine for turning out pepsin from blood and liken it to a machine for turning out (say) newspapers, but the secretion of pepsin is not mechanical, nor is the output of newspapers vital.”

The strangest point about the materialistic or mechanistic conception of things is that every machine without exception has an inherent purpose and design. It is intended by its inventor to do certain definite things in a certain definite way. If an animal, therefore, is a machine, it must have a design and

purpose, and presumably a designer. But the main idea underlying the mechanistic hypothesis was to eliminate the idea of purpose altogether, and reduce the universe to an accidental configuration of lifeless atoms.

It is difficult to see how this attempt to reverse the rôle of Potter and Pot could ever have satisfied enquiring and well-balanced minds, but it is a fact that mechanistic views of life, after a period of almost general acceptance, are still prevalent among biologists whose education in the principles of logic has been somewhat hurried.

The majority, however, have returned to the saner view that it is useless to explain the known by the less known. Human purpose is a primary fact of experience, and the embodiment of a human purpose in a tool or machine is a process which cannot be denied on any reasonable grounds. Now, a “purpose” is a datum of the mind, which cannot be reduced to any simpler elements. The purpose of a machine is the psychical element embodied in it. Briefly, we may call it the “soul of the machine.” Every machine has a psychical element, a purpose, a “soul.” It is therefore, simpler, more in accordance with sound philosophical principles, more direct and “economical” to explain machines in terms of psychology than to explain human bodies in terms of mechanism.

The victories of Hephæstus are victories of mind over matter. The “mechanical age,” which to some appears as the very negation of the soul, is, on the contrary, the age of supreme psychical achievement.

Science and invention are for ever annexing fresh regions of the universe and subjecting them to the free play of our mental faculties. The process of bringing material things into subjection to our will is a process of sublimation, which does not drag us down to the dust, but raises up dust into the realms of immortal spirit.

III

... The liquid Ore he dreind
Into fit moulds prepar'd; from which he form'd
First his own Tool; then, what might be wrought
Fusil or grav'n in mettle.

At the time when the first man decided to go forth into the world provided with weapons not furnished by nature (in other words, “unnatural”) the lines of development adopted by nature had ended in an *impasse*. Mere size had been found ineffective, and the giant Reptilians had disappeared from the earth. The Mammoth and the great Rorqual Whale were the largest animals then in existence, and nature had retraced her steps somewhat as shipbuilding did in the nineteenth century after the building of the *Great Eastern*. Of the countless forms of animal and vegetable life, many had disappeared entirely, being no longer suited to climatic or other changes of environment. There was no prospect of higher development unless an entirely new path could be found. We may put the situation in another way by paraphrasing an ancient tribal account of the origin of things:

“In the Beginning there was the Sun and the Earth.

“And the Sun and the Earth said: Let there be Life. And the Earth covered itself with a living coat of green, fed by the Sun.

“And the Earth said: Let there be Moving Life. And Life began to Move about, fed by the Life that was green and stood still.

“And the Earth said: Let there be Man, and let him be fed by the Green Life and the Moving Life, and let him subdue all Life, and let him subdue Me and serve Me and make Me great in the Heavens.”

And so the Earth brought forth Man, her latest and greatest Experiment. For a long time he was a rather inferior animal, but when he began to throw stones and spears he launched out on his true career, a career destined to

culminate in the complete mastery of his native planet and the apotheosis of the Earth.

The pre-human Animal had already learned to use the world of plants for purposes of nutrition, and to use the mineral world for dwelling purposes. Improving upon the methods of his predecessors, man made the land and sea his province, and drew from it not only nourishment but the means of extending his dominion. "He that hath, to him shall be given." And this extension became more and more rapid. Living, as we do, in an age of continually accelerated progress, we find it difficult to realize its rapidity.

We are caught in a flood. From day to day things are changing. What we write to-day is obsolete to-morrow. The clock ticks on the mantelpiece as it did twenty years ago, but outside in the road is the roar of motor traffic. The clip-clop of the horses' hooves is no longer heard. Hephæstus has put his fires into the interior of the motor engine, and found yet another way of using sunlight accumulated in the earth millions of years ago, instead of relying upon the solar energy stored in the grass eaten by the horse.

Fire has made all things new. We are surrounded by its gifts. My pen has passed through many fires before it reached my hand. All round me are traces of machinery and mechanism. The paper on which I write is calendered in the paper mills. I am aware that it is not as lasting as "hand-made" paper, and that it will not survive centuries of use like Gutenberg's bibles. But what matter? If there is anything worthy of survival in what I write it will survive, even though it may require to be cast in bronze. If it is good it will be *aere perennius*.

Mankind may be likened to a vast army on the march. It is preceded by:

(1) Pathfinders, called explorers, inventors, and discoverers, whose business it is to find new avenues of development and achievement.

Then come the

(2) Commanders with their staff of Organizers. These direct and co-ordinate the movements of the masses in directions judged to be for the general welfare. They are followed by

(3) The Rank and File, consisting of those who live and work along accepted and well-established lines in all grades of society, taking no risks and making no changes.

(4) The Stragglers, or those who work for themselves alone, without reference to the needs and prospects of the community. Among these must be classed the squatters and backwoodsmen, and small crofters and peasants who grow their own food and weave their own cloth. They do not belong to civilized society, though they do no harm to it and cannot quite escape its influence.

(5) The Campfollowers and Vultures, who have no regard for the welfare of the community, but prey upon it for their own ends.

Classes (1) and (2) are by far the most important and valuable constituents of the human race. They are the growing element, the “cambium,” the grey matter of the brain. Whoever has sufficient originality to strike out an original course, combined with loyalty to humanity at large or to a smaller community; whoever is capable of leadership, whether in war or peace, art or commerce, industry or politics; whoever can lead others forward and inspire them with courage to face difficulties—he belongs to the *élite* of mankind, to whatever grade of society he may belong. Far behind his class come the rank and file of commonplace drudges, who work in a rut and submit to being led like sheep. They may be clerks, domestic servants, trade union operatives, pensioners, or small investors. They form the large, undistinguished, but useful mass of humanity. Many kings have belonged to this class.

Both the large capitalist and the trade union boss I should class among the Commanders, and I should assign them a high rank in human progressive elements. The former is often a Pathfinder in commerce and industry, and the latter often points the way to the betterment of manual workers. Both are in a position of great power, but are exposed to the temptation to abuse it. The financier may succeed in restricting the free market in an important human community for his own enrichment and aggrandizement. The trade union boss may make a “corner” in a certain form of labour and so deprive the community of some essential commodity, such as housing accommodation. When this degeneration takes place, both these types must be put into the Vulture class.

The same judgment must be passed on those who use the machinery of the medium of exchange to further objects contrary to the interests of the

community, such as usurers, and purveyors of intoxicant drugs. These also are among the Vultures.

When the first armed man transferred a method employed by nature into a new medium, and derived his armour from the outer world instead of growing it in his own organism, he took a step which led to many similar re-interpretations. One of the most important of such steps was Organization. The very word recalls its “organic” origin. The organs of the human body form an interdependent community with a common interest. The organs again are composed of tissues, and the tissues of individual cells having, as many biologists believe, a rudimentary consciousness of their own. All these—some millions of millions of millions—form a vast and closely-organized community consisting of many more distinct individuals than there are human beings on earth. The human race is such an organized community. A swarm of bees is another, but while the latter only deals with “natural” food-supplies and housing materials, the human community, thanks primarily to the use of fire, gathers its resources from realms utterly inaccessible to the ordinary animal, and establishes an unassailable superiority. Thus the human army, consisting of Pathfinders, Organizers, Rank and File, and Stragglers, with a trail of Vultures behind, presses ever forward on its victorious march of progress.

Its general procedure may be represented as follows:

The Pathfinders are in front, seeking out new avenues of advance. They may discover a new coal mine or oil well or mineral deposit; a more economical method of lighting and heating; an improved method of weaving or printing; a new medicine; a new formula for expressing numerical relations; an improved method of transmitting news; or merely a simplified method of mending socks. Whatever it may be, the new discovery is passed on to the organizers, the captains of industry, the capitalists and financiers, and the trade unions. In a well-organized community, the discovery or invention is given every opportunity of proving its value. Where vested interests and monopolies stand in the way, either in the camp of the capitalists and property holders or in the ranks of labour, much opposition may be encountered, and the community may be deprived of the advantages of the new discovery. But if there is no such opposition, the work incidental to the utilization of the discovery is distributed by the organizers among the rank and file, consisting of

mechanics, clerks, and small investors. As soon as the industry is successfully established, the Vultures begin to hover round. Some of them seek to drive the industry into a corner where it can only exist by serving the interests of the Vultures. Other Vultures endeavour to corner the labour trained by the pathfinders and pioneers and hold the new industry to ransom. But in a well-organized community these nefarious activities are kept within bounds. The pathfinders, the organizers, and the rank and file are given their due credit and reward, and the community reaps the full benefit of the discovery.

And now let us examine the activities of the Pathfinders. In classical times the most audacious and renowned of these were the Phœnicians, who, armed with their shields and corselets of “oak and triple bronze,” sailed through the Pillars of Hercules out from the tideless Mediterranean into the unknown terrors of the Atlantic. Their ships were seen in the Baltic, trading woven purple garments for amber, and on the British coasts in search of tin. At the request of an Egyptian Pharaoh, they circumnavigated Africa, and brought back wildly improbable but, nevertheless, true stories about new constellations and the sun culminating in the north.

In Egypt itself explorers and discoverers of another kind were busy. The science of Chemistry was born there, and named after “Chem,” the native name of Egypt. In Greece and its colonies the science of Geometry attained a high standard, while Syracuse stands out as the home of Archimedes, one of the greatest mathematicians, physicists, and inventors of all time. These Pathfinders enriched humanity with priceless gifts. They surveyed the field of possible discovery from a high altitude, and their clear vision traced out the paths to higher achievement, to be trodden by their successors.

Had they lived in better-organized communities, their labours would have been turned to greater advantage, and would have benefited a greater number of their fellow men. But the day of capitalism and mass-production had not yet come. The splendid achievements of an Archimedes only served to benefit a tyrant, and his single-handed scientific fight against the Romans ended in disaster to his beloved city.

In Greek and Roman days, owing to the lop-sided organization of human society, scientific discoveries could only benefit a few powerful people. The downtrodden underlings were unaffected by the work of the Pathfinders.

They had to await the dawn of the Industrial Age before they could take part in the progress of the *élite*. Yet it must be put to the credit of the Hieros, the Medicis, and the greater of the Bourbon and Stuart Kings that they fostered the advancement of science and protected the Pathfinders from the persecution of ecclesiastical and other vultures.

The invention of gunpowder laid low the chivalry of the Middle Ages. The Hobbling God triumphed over the mail-clad monopolist on horseback. The Sudden Fire armed the foot-soldier with a winged shaft of death. The stone-thrower had become the unerring sharp-shooter. The range of human power increased apace. The glass-blower with his fiery furnace produced cunning contrivances of glass which increased the range of the human eye twenty or thirty-fold. A free philosopher like Spinoza could discourse of the fundamental substance without fear or favour, since the grinding of optical glasses assured him an independent livelihood. Then came the French Revolution with its clarion call to a wider organization of humanity, which was eventually brought about, not by speeches and pamphlets, but by the harnessing of coal to inaugurate the Mechanical Age.

The nineteenth century saw the fulfilment of the dream of an organized Humanity, a human world unified, not by the spread of humanitarian philosophies, but by the material bonds of progress.

Some of my critics have ridiculed my “robust” belief in the reality of progress through invention and the use of machinery. I see no reason to modify my view. A generation ago, Ruskin deplored the spread of railways which scored their lines through “bleeding landscapes.” To me a railway line is a thing of beauty, wherever it may be found. It is a symbol of a higher will and of a purpose transcending the puny sphere of the individual. To see the rails is as if the sinews and muscles of some supernatural being were made visible and accessible to me. And who shall say that I am wrong in that feeling? A railway is an organization superior to man. “Societies,” says Edward Carpenter,^[3] “not only of bees (as Maeterlinck has shown), but of all creatures up to man, have, *qua* societies, a life of their own, inclusive of and superadded to that of their individual members.” And so a railway appears to me as an individual of a higher order, closely organized and endowed with a life of its own. Its mental equipment is its personnel, its directors, clerks, drivers, guards and porters. Its body consists of rails, bridges, tunnels, stations and rolling stock. It has a soul and a living

purpose. It has a power of self-maintenance and self-preservation and a rudimentary memory.

[3] *The Art of Creation.*

I heartily sympathize with the child who adores a locomotive engine. To him, nothing could be more lavishly fraught with beauty. I saw a little while ago a powerful engine at Waterloo Station at the head of an express train for Bournemouth. It was a thing of perfect and satisfying beauty. Its lines suggested calmness and strength and speed. The wisp of steam about its safety valve spoke of well-controlled powers within its massive frame. The driver, a clear-eyed man with trusty hands and sturdy body, was a noble representative of Hephæstus. One felt that the load of health-seeking humanity behind him was in safe hands, that his nerves and muscles would work as perfectly as the engine he regarded with such loving attention, and that the 108 miles of the non-stop journey would be accomplished without a hitch. The beneficent monster lay there in the morning sun, ready to spring forward on its swift trajectory across the “bleeding landscape”! And there are people who can look upon the eager children in the train, armed with their spades and buckets for playing in the health-giving southern sunshine, and deny the progressiveness of the Mechanical Age!

They will probably argue, of course, that if it were not for the smoke of the city, due to the “fires of Hephæstus,” there would be no necessity to have trains to carry the children to the seaside. But the smoke of cities is due mainly to incomplete combustion in domestic hearths, and the remedy lies not in the discouragement of mechanical invention but in its further extension until it provides us with a smokeless city.

The fires of Hephæstus are fashioning a new world. They are welding humanity into a coherent mass. All the metals, all the ninety chemical elements, are being pressed into service. Whose service? The service of a race whose destiny we can as yet only dimly appreciate. Already we command temperatures varying from a region within a few degrees of the absolute zero to within a few hundred degrees of the heat of the sun. The sight of our eyes has been supplemented to such an extent that we can appreciate and deal with some fifteen octaves of visible and invisible light instead of the single octave “naturally” accessible to our sight. We command pressures of tons per square millimetre and degrees of vacuum down to a hundred-millionth of an atmosphere. We can photograph the

track of a single atom tearing its way through moist air. We can print 500,000 copies of a paper of 150,000 words daily and sell it for a penny. We have banished bears and wolves from our home countries and have learnt to wage war against invisible germs of disease. We have acquired the power of bringing beautiful music into the homes of our humblest citizens. The luxuries of our forbears are the common possessions of our own generation. If we are not happier than our ancestors, the fault lies in ourselves, in our ingratitude and lack of imagination. Or must we conclude that happiness is a negligible thing in the great scheme of progress, and that that scheme does not concern itself with our individual feelings?

I for one believe that happiness is on the up-grade, too. Hephæstus is not only a strong and a clever god, but a god with a sense of humour and a very lovable character.

IV

Forerun thy peers, thy time, and let
Thy feet, millenniums hence be set
In midst of knowledge, dreamt not yet.

The empire of Hephæstus is expanding before our eyes. His bellows blow through our blast furnaces. His anvils ring in thousands of factories. His engines career over the land, and his steamers over the sea. His internal-combustion cars are strung out along the roadways, and his avions run their furrows through the clouds. Wherever he goes, he lightens the burdens of humanity and gladdens all hearts with his gifts. He brings warmth and light into dark places. He draws mankind closer and closer together. He draws them together without diminishing the distance between them. He simply makes that distance impotent to hold them apart. He enables the city children to leave their slums and gather fresh health in the open country. He gives the grimed city worker fresh clear air wherein to sleep. His lorries, thundering along the highways, tighten the bonds between the farmer who gathers the fruit and the town dweller who consumes it.

Nor is Hephæstus satisfied yet. His task is but half finished. The whole earth must be Vulcanized.

The God of Fire and of Iron hobbles over the broad earth, seeking and finding new paths of advancement towards a better and fuller life.

Man has become less and less limited by his permanent organism, that Body which in the beginning was all he possessed. He has now as many different shapes as ever Proteus had, and he can assume any of them at will. At one time he has the shape of a rowing boat, his arms prolonged seven feet or more till his blades skim the water. Again, his legs will be two wheels, and his body will be like those steeds with fire-breathing nostrils which Hephæstus made for the King of Colchis. Again, the pupil of his eye will grow until its diameter measures 100 inches and then he will sound the depths of the heavens with the power of 100,000 eyes in one. When he goes to war, he throws his "stone" some seventy miles and arms it with vast

destruction. He belches forth deadly vapours which choke and strangle his enemies. He becomes an armed slug creeping over all obstacles and destroying as he creeps.

In times of peace he dresses in various garbs to suit his changing occupation. His womenfolk emphasize and glorify their variety of personal presentation and expression. Protea shines in all the colours and shades of the rainbow, and invents new colours and new forms every day, forms of dress, of outline, and of her very bodily figure.

The resignations and renunciations of former ages are forgotten. Music, travel, pictures, education—the luxuries of past generations are the common property of all to-day. That is the sort of Communism which is feasible and reasonable. Take all the luxuries and good things of this life and bring them within reach of all by cheapening their cost of production. Travel at a pound a mile is a luxury. Travel at a penny a mile becomes a “necessity” as soon as people get used to it and take it for granted.

Security and seclusion are privileges cherished by what used to be called the “aristocrats.” Place both within reach of the masses by insurance and by transport facilities. Make all these luxuries and privileges common property. It is not so many centuries ago that a mariner’s compass was bartered for diamonds. To-day it costs a penny. And then we come upon a difficulty. When a thing becomes common property, or so cheap as to be of practically no price, it ceases to be coveted. There is a tendency deep-rooted in the human heart to reach out beyond the things of to-day for something not yet attained. When Communism has shared out the earth, it will ask for the moon. Humanity will never assume a “dead level of mediocrity.” There will always be the Thyroid Type, energetic, inquisitive, sensitive, inventive, and restless. The class of Pathfinders will be recruited from that type. The Pituitary Type will furnish commanders and organizers, while the Adrenal Type will furnish the rank and file. But the intermixture of the various types will bring out special constitutions and varieties in unexpected quarters, and it behoves us to keep “*la carrière ouverte aux talents.*”

Thus the human army will march forward, led by a Pillar of Fire. Its pace will be constantly accelerated, but we shall hardly be aware of it, for our methods of measuring time will change in the same ratio. A well-made motor-car on a smooth road can hum along sweetly at sixty miles an hour,

and its occupants will be less impressed and excited than they would be by the horse careering at 20 miles per hour! And what are all these speeds to the absolutely smooth planetary speed of nineteen miles per second!

It has been asserted that the tremendous powers conferred upon man by machinery have produced the terrific wars of recent times by placing power in the wrong hands. There may be some justice in this, but the amount of destruction is negligible in comparison with the amount of construction and reparation. When Kitchener saved a few hundred casualties at Paardeberg by sparing his men, he lost 10,000 at Bloemfontein by enteric fever. The bloodiest war is sometimes the least costly on balance. The greatest war of all history ended but seven years ago, and already the beaten nations have healed up their wounds and are looking forward to the future with greater confidence than many of the “victorious” nations. Mankind is increasing the rapidity of its nervous reaction to emergencies and injuries. The leucocytes of the body politic set to work sooner and more effectively to heal the damage. Mankind as an organism is daily improving its circulatory system, *alias* its transport facilities. Its nervous system closely imitates organic nerves by its cables and land-lines, and it improves upon nature by adding wireless communication and resonance to the organic devices of the body.

Mankind is being moulded into a single compact self-contained body on the anvil of Hephæstus. It is still subject to feverish ailments and some chronic diseases. But the general outlines of future development are clearly discernible. There must be pauses and set-backs. The brilliant scientific and artistic development of classical Hellenism was all but lost in the destruction of the Roman civilization. But the Arabs kept the flickering flame alive until Europe awoke from its torpor and lit its torch once more. Since then there has been no turning back. The age of science, discovery, and invention, the age of mechanism and machinery and power, has come and come to stay. Man, liberated from mechanical drudgery by the machine, has time to develop his intellectual and artistic powers. His necessities being supplied by pressing a button, he is liberated to enjoy a more varied existence. If some men and women are still bound to monotonous tasks, it does not necessarily mean unhappiness, for the Pathfinders’ life is one of constant care and much anxiety. Discovery is 90 per cent. failure, and something like a daily routine of regular and monotonous work sometimes appears in the light of a blissful refuge. Besides, the human heart and

organism easily falls into a routine in which daily work is hardly felt, being done without conscious effort. This fact is often forgotten by those who envy the life of the organizer and financier, which is one of much risk and anxiety, relieved by an occasional big prize.

The ideal state of things would be attained if those of an adventurous disposition could be given the adventurous part of human activity, and if those who are plodders by nature could be left to do the plodding.

That is a matter for future development, either consciously fostered by wise leaders of our race, or unconsciously evolved from the depths of wisdom hidden below the threshold of the racial consciousness.

But the goal is in sight. The earth is being organized and unified under the ægis of the human race, the protoplasm of this planet, the race which, transcending the mechanism and long-established traditions of its own germ-plasm, enlarged and multiplied its functions until it acquired the use of, fire. Upon that achievement it built an unprecedented form of life, a super-“natural” edifice of infinite power, as yet but dimly realized, but which in its full beauty and perfection will be nothing less than Divine.

THE END.

Transcriber's Notes:

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Punctuation and spelling inaccuracies were silently corrected.

Archaic and variable spelling has been preserved.

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*** END OF THE PROJECT GUTENBERG EBOOK HEPHÆSTUS ***

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